Mandate

Insuring the agents that spend your money.



By Lewis Wilson. October 2026.

Mandate is a theoretical model for research and education. It is not an insurance product, quote or financial advice. Landscape statements are as of October 2026. Every parameter is an illustrative assumption. Legal summaries are not legal advice.

Executive summary

Software can now spend money on our behalf. AI agents compare products, fill baskets, book trips and pay supplier invoices, and the payment networks have spent the last eighteen months building the rails to let them check out [M1, M2]. What has not been built is an answer to a simple question: when an agent gets it wrong, who pays?

Today the answer is unclear. The card networks have launched agent frameworks, but liability beyond straightforward fraud remains largely unresolved [M4]. Under UK payment law, a payment is authorised if the payer consented in the agreed way, so a mistake made by an agent you set up may still count as your authorised payment [M6]. Specialist AI insurance exists, but it is written for businesses deploying AI, not for the people relying on it [M10].

This paper designs a theoretical product to fill that gap. **Mandate** insures people and small businesses against losses caused by AI agents making payments: wrong items, wrong prices, duplicated orders, manipulated agents and counterfeit merchants. It sets out what the cover would include, how claims would interact with existing protections, and, in full, how it would be priced.

The pricing model is deliberately transparent. Premiums are built from the frequency and severity of six types of loss, adjusted by factors the policyholder controls: how much autonomy the agent has, how payments are authorised, which payment rail is used, and what safeguards are switched on. That transparency is the point. If safer configurations are cheaper to insure, insurance becomes a market signal that rewards safer AI.

The model also confronts the feature that makes AI risk unusual: errors are systematic. A single faulty model update can affect thousands of users at once, so an insurer must price correlated losses, not just individual ones. The paper shows how a simple common shock changes the capital an insurer would need.

1 Introduction: when software holds the card

Delegation is not new. People have always asked others to buy things for them, and the law has long understood agency: if you authorise someone to act for you, their actions bind you within the limits of that authority. What is new is the agent. An AI system can make thousands of decisions a day, at a scale and speed no human assistant could match, and it makes mistakes differently from a person: confidently, repeatedly, and sometimes because someone has deliberately fed it misleading instructions.

The infrastructure for agent payments is arriving quickly. Mastercard launched Agent Pay in April 2025, and Santander and Mastercard completed what they described as Europe's first live end-to-end payment executed by an AI agent, on live payments infrastructure in a controlled pilot, in March 2026 [M1, M2]. Protocols now exist to record what a user actually intended: Google's AP2 uses cryptographic "mandates", signed records of the user's instruction, while OpenAI and Stripe's protocol issues payment tokens scoped to one merchant, capped at a maximum amount and time-limited [M3].

Consumers are not yet convinced. In an April 2026 survey of online adults in the US, UK and Canada, three-quarters said they were uncomfortable with an agent paying autonomously, even with preset spending limits, and they expected the AI platform to be responsible when something went wrong [M21]. Trust, rather than technology, has been identified as the main barrier to agentic commerce [M3].

Insurance has historically been one of the ways societies make new risks acceptable. Motor insurance made cars usable by ordinary people; cyber insurance has shaped how companies secure their systems. This paper asks whether insurance could do the same for agentic payments, and what such a product would need to look like.

2 How agents pay

An agent payment passes through several layers, each with its own failure points (adapted from [M3]):

1. **Discovery:** the agent searches, compares and selects a product or service
2. **Checkout:** the agent builds a basket and passes it to the merchant
3. **Authorisation:** the agent proves it has permission to pay. This may be a signed mandate, a scoped token, or simply a stored card
4. **Settlement:** money moves over a card network, bank transfer or alternative rail

The strength of the authorisation layer matters enormously for insurance. A signed mandate records exactly what the user asked for, so a dispute can compare the instruction with the outcome. A browser agent using a saved card leaves far weaker evidence. The payment rail matters too, because each rail carries different statutory and scheme protections (Section 4).

3 How agents fail

Agent errors fall into recognisable patterns.

Misinterpretation. The agent buys the wrong size, date, model or quantity. In one shopping-agent benchmark, the most common failure (39.1% of errors) was

asserting product attributes that had never been verified [M15].

Price error. Agents can be confidently wrong about prices. One study of AI shopping assistants reportedly found that when prices were wrong, they were wrong by a median of $300 [M18].

Duplication. Retries and timeouts can produce the same order twice.

Manipulation. Agents read web pages, product descriptions and documents, and those can contain instructions aimed at them. In one test, a product description told agents to add an extra item to every order; agents added the item in 40% of sessions, and told the customer it was required [M16].

Counterfeit merchants. Researchers have shown an agentic browser checking out on a counterfeit store using saved card details [M17].

Scope creep. An agent instructed to buy one thing may also sign up to a subscription or add services.

Business payments. For businesses, the stakes are larger. Payments fraud is already widespread: 76% of US organisations experienced attempted or actual payments fraud in 2025, and vendor impersonation is a rising business email compromise tactic [M19]. Security researchers warn that attackers can adapt these techniques to target agents that hold delegated authority to approve payments [M20].

One feature of AI failure matters especially for insurance: it is systematic. A model that makes an error will tend to repeat it until it is corrected [M13]. Human mistakes are largely independent; agent mistakes can be correlated across thousands of users.

4 Who pays today

UK law allocates payment losses through a patchwork of rules that were designed for human payers.

Consent and authorisation. Under the Payment Services Regulations 2017, a payment is authorised if the payer consented in the form agreed with their payment provider [M6]. If you set up an agent and gave it your payment credentials, its payments may well be authorised, even when the outcome is not what you intended. A commentary on the Oxford Business Law Blog describes the result as a potential liability gap between consumer protection, payment law and product responsibility [M6].

Payment rails. Protection depends heavily on how the payment was made [M6]. Credit card purchases benefit from chargeback rights and, for items priced over

£100 and up to £30,000, the card issuer's joint liability under section 75 of the Consumer Credit Act [M22]. Debit cards offer chargebacks but not section 75. Bank transfers offer little protection for errors. Stablecoin rails typically lack chargeback infrastructure altogether [M7].

Fraud reimbursement. Since October 2024, victims of authorised push payment fraud over Faster Payments and CHAPS are entitled to reimbursement of up to £85,000 per claim [M8]. That helps where an agent is tricked into paying a fraudster, but it does not cover an agent that simply buys the wrong thing.

Businesses deploying agents. The CMA's March 2026 guidance makes clear that consumer law applies in the same way whether a customer deals with a human or an AI agent, and that businesses are responsible for what their agents do [M9]. That protects consumers dealing with a retailer's agent. It does less for consumers whose own agent makes the mistake.

The networks. Visa, Mastercard and American Express have launched agent frameworks, but liability between merchant, issuer and agent platform remains largely undecided beyond straightforward fraud [M4]. American Express is reported to have the clearest commitment so far [M5].

The result is a matrix of partial protection. A consumer whose agent makes a mistake on a credit card has a reasonable chance of recovery; the same mistake on a bank transfer or a stablecoin wallet may leave them with nothing.

Who pays? A wrong-size trainers claim

Wrong size trainers | Duplicate concert tickets | A fake store | A prompt-injected add-on | A tampered supplier invoice | Wrong non-refundable flights

Wrong size trainers, £95.00. Asked for size 9, the agent bought size 11 in a final-sale promotion that cannot be returned.

On a debit card, without Mandate you would expect to bear £57.00 of this £95.00 loss. With Mandate you bear £18.50, and Mandate pays you first, then pursues the recovery.



Expected shares of one illustrative loss. Recovery chance on this rail for this kind of loss: 40% (merchant or card issuer (chargeback)). Agent platforms have no settled liability today, shown as £0. Mandate terms: £10.00 excess, 10% co-payment. Not legal advice.

Who pays the £95.00 loss, in expectation (illustrative)

Payer	Without Mandate	With Mandate
Merchant or card issuer (chargeback)	£38.00	£26.01
Agent platform	£0.00	£0.00
Mandate (net of recoveries)	£0.00	£50.49
You (excess, co-payment, fault, limits)	£57.00	£18.50

5 Why insurance, and why now

Insurance markets for AI are forming, but from the supply side. Munich Re has insured the performance of AI models since 2018, pricing cover on the robustness

of the model [M12]. Armilla, which describes itself as the first Lloyd's coverholder dedicated exclusively to AI liability (2024), now covers AI agent failures with limits up to $25 million per organisation [M10]. Traditional policies remain thin on AI: one broker noted that a typical $5 million technology policy might allocate only $25,000 to AI-related liabilities [M11].

These products serve companies that build or deploy AI. No mainstream product protects the individual or small business on the receiving end of their own agent's mistake. That is the gap Mandate is designed for.

Insurance could do more than reimburse losses. Risk-based pricing communicates risk. If a policyholder can see that requiring approval above £50, or using a payment rail with chargeback rights, reduces their premium, the premium becomes a running guide to safer behaviour. If insurers pay claims and then recover from merchants, platforms and networks, they create pressure further up the chain for safer agents. This is the role insurance has played in other emerging risks, and it is the central argument of this paper.

6 Designing the cover

Mandate covers six insured events for individuals (misinterpretation, price error, duplication, manipulation, counterfeit merchants and scope creep) and adds invoice manipulation, payee substitution and overpayment for businesses (Section 4 of the specification).

4.1 What is covered: people

In plain English

- **Misinterpretation:** the wrong item, size, date or quantity.
- **Price error:** paying well above your limit or the quoted price.
- **Duplication:** the same order placed twice.
- **Manipulation:** the agent is prompt-injected into adding items or changing details.
- **Counterfeit merchant:** the agent pays a fake store.
- **Scope creep:** a subscription or extra service you did not ask for.
- **Booking failure** (optional add-on): non-refundable travel booked wrongly, with limited consequential cover.

Illustrative policy wording

The Insurer will indemnify the Insured against Financial Loss arising from a Payment executed by an Authorised Agent where the Payment departs from the Insured's Instruction by reason of a Covered Peril, as set out in the Schedule.

4.2 What is covered: businesses

In plain English

Everything in 4.1, plus an agent paying a tampered or fake invoice, paying a supplier whose bank details were changed by a fraudster, and overpaying an invoice.

Illustrative policy wording

For a Business Policy, Covered Perils further include Invoice Manipulation, Payee Substitution and Overpayment, each as defined.

4.3

What you pay first, and how claims are settled

In plain English

- **Excess:** you pay the first part of each claim.
- **Co-payment:** you keep a share of each loss above the excess.
- **Limits:** there is a maximum per claim and per year.
- **Recovery first:** Mandate pays you promptly, then pursues the merchant, platform, card scheme or bank itself.
- **Evidence:** a signed mandate or agent log gets a claim paid faster, with no shared-fault reduction.
- **Shared fault:** the payout is reduced if you dismissed the agent's confirmation request or switched off required controls.

Illustrative policy wording

Each Claim is subject to the Excess, the Co-payment and the Limits stated in the Schedule. On payment the Insurer is subrogated to the Insured's rights of recovery. Where the Insured dismissed a confirmation request or disabled a Required Control, the Insurer may reduce the amount payable in proportion to the Insured's contribution to the Loss.

4.4

What is not covered

In plain English

- Changing your mind when the agent did what you asked.
- Purchases outside the set-up you declared, for example with caps switched off.
- Losses already paid back another way.
- Deliberate misuse.
- Crypto rails, unless you add them.

Illustrative policy wording

The Insurer shall not be liable for (a) any Payment that conforms to the Instruction; (b) any Payment made outside the Declared Configuration; (c) any Loss already reimbursed from another source; (d) any deliberate act of the Insured; or (e) any Payment on a crypto-asset rail, unless the Crypto Rail Extension is shown in the Schedule.

4.5

What Mandate requires

In plain English

- A spending cap on every transaction.
- Agent logs kept for 90 days.
- Telling Mandate promptly about a suspected loss, for example within 30 days.

Illustrative policy wording

It is a condition precedent to liability that the Insured maintain a per-transaction Spending Cap, retain Agent Logs for not less than ninety days, and notify the Insurer of any suspected Loss within thirty days of discovery.

Three design choices do most of the work.

Pay first, recover later. Policyholders should not have to navigate chargebacks, section 75 claims and platform complaints before being paid. Mandate pays promptly and then pursues recoveries, so its net cost depends on the payment rail.

Evidence reduces friction. Claims supported by a signed mandate or agent log can be assessed by comparing instruction and outcome. These claims are paid faster and without any shared-fault reduction. This creates a reason to choose agents that keep good records.

Shared responsibility, not blame. An excess, a modest co-payment and a shared-fault reduction (for example, where the policyholder dismissed a confirmation request) keep the policyholder engaged without undermining the cover.

Exclusions are designed to keep the product insurable: buyer's remorse where the agent followed instructions correctly, purchases outside the agreed configuration, and deliberate misuse.

7 Pricing the risk

Mandate is priced with a standard collective risk model, made transparent.

The rating ledger: autonomous power user

Lets a browser agent buy freely on the open web with a saved debit card. Every figure is illustrative. The web version lets you change every input.

POLICY SCHEDULE ILLUSTRATIVE



Mandate Personal

Policy no. MND-2026-774430

Mandate Personal for a person whose agent shops for them: about £9.41 a month, for an expected £116.90 a year in claim payments.

Insured	You (an illustrative policyholder)
Configuration	60 purchases a month, average £45.00; fully autonomous; browser agent with saved card; reliability score 60
Spending cap	£500.00 per transaction (largest purchase £600.00)
Payment rail	Debit card
Excess	£10.00 per claim
Co-payment	10% of each loss above the excess
Limits	£1,000 per claim, £2,500 a year
Claim evidence	Signed mandate or full agent logs: no shared-fault reduction

Monthly premium	**£9.41**
Annual premium	£112.92
Expected annual claim payments	£116.90

= pure premium £75.45 + risk margin £5.29 + expenses £32.18

Rating ledger

Every multiplier, line by line. Rates, factors and severities are illustrative assumptions. Open a peril to see its full build-up.

	Agent purchases a year (60 a month × 12)		720
	› **Misinterpretation**	4.072/yr	£43.33
	› **Price error**	1.454/yr	£5.17
	› **Duplication**	0.713/yr	£14.82
	› **Manipulation**	0.386/yr	£2.57
	› **Counterfeit merchant or invoice fraud**	0.272/yr	£4.57
	› **Scope creep**	0.907/yr	£5.00
=	**Pure premium P0 (expected net cost, sum of perils £75.45)**		£75.45
+	Risk margin: 0.1 × standard deviation £52.93		£80.74
+	Expenses: £12.00 fixed + 25% of (P0 + risk margin)	£32.18	£112.92
=	**Annual premium**		**£112.92**
/	Monthly (annual / 12)		£9.41

Peril rows show expected events a year and expected net cost a year. Figures are rounded; totals use unrounded values.

Frequency. For each type of loss, the expected number of events in a year is the number of agent transactions multiplied by a base probability that a transaction produces an unrecovered problem, adjusted by rating factors. More autonomy, weaker authorisation, an open web of merchants and weak defences against manipulation all increase frequency. Approvals, signed mandates, merchant allowlists and good defences reduce it.

Severity. The size of each loss is modelled with a lognormal distribution, a skewed shape that captures many small losses and a few large ones, scaled to the policyholder's typical purchase and capped by their spending limit. Spending caps therefore act directly on severity: the single most effective safeguard in the model.

Recoveries. Expected recoveries depend on the payment rail. Credit cards recover a large share of losses; stablecoins recover none. Because Mandate pays first, recoveries reduce its net cost rather than the policyholder's payout.

Payout. Each loss passes through the excess, co-payment, per-claim limit and any shared-fault reduction, and annual payouts are capped.

Premium. The pure premium is the expected net cost. To it are added a risk margin proportional to the variability of losses, and expenses. Claim-free years reduce the premium through a bonus-malus scale.

Every parameter in the model is illustrative. No insurer has yet published loss data for agentic payments, and the paper does not pretend otherwise. The value of the model lies in its structure: it shows which factors drive the risk, how they interact, and how much each safeguard is worth.

Three policyholders compared

The web version simulates 10,000 policy years for any configuration. Here, the analytic results for three presets.

Illustrative annual figures from the model

Policyholder	Annual premium	Expected payout	Chance of at least one loss
Cautious shopper	£24.00	£0.69	8%
Autonomous power user	£112.92	£116.90	100%
Small business, invoice agent	£1,561	£1,149	87%

8 The systemic problem

Most personal insurance relies on independence: one person's burst pipe says little about another's. AI agent losses break this assumption. AI errors recur until they are corrected [M13]. Agents built on the same model share the same weaknesses, so a single faulty update or a widely circulated manipulation technique could trigger losses across thousands of policies at once.

The model captures this with a common shock: a small annual probability of a systemic event that multiplies error rates across the whole portfolio for a short period. Even a modest shock dramatically widens the tail of the insurer's loss distribution and increases the capital needed to remain solvent.

Insurer view

The web version simulates a 10,000-policy portfolio with and without the common shock and reports the capital an insurer would need (99.5% value at risk). See Methodology for the model.

This has practical consequences. A real Mandate would need aggregation limits, clear event definitions (for example, treating losses from one model version as one event), reinsurance, and pricing that reflects concentration across agent providers. It would also have a strong interest in monitoring agent providers and in recovering systemic losses from them. Cyber insurance offers the closest precedent, and its early history suggests that insurers who ignore correlation learn about it expensively.

9 Businesses

For businesses, the dominant risk is not a wrong pair of trainers but a wrong payee. Agents that read invoices, match them to orders and pay suppliers create a single point of failure that attackers are expected to target [M19, M20]. The business version of Mandate therefore prices controls familiar to any finance team: segregation of duties, verification of changed bank details, and e-invoicing, each of which sharply reduces modelled fraud frequency. The lesson matches long-standing finance practice: automation is safest when it inherits good controls, not when it replaces them.

10 Limitations and open questions

- **Data.** Parameters are illustrative. Real pricing would require loss data that does not yet exist publicly.
- **Law.** The legal map is a summary and is moving quickly; network liability rules in particular are still being written [M4].
- **Moral hazard and adverse selection.** Pricing and evidence requirements reduce, but do not eliminate, these classic problems.
- **Regulation.** A real product would require authorisation and would sit under the FCA's Consumer Duty.
- **Who should insure.** It is possible that the natural home for this cover is the payment network or agent platform rather than a standalone insurer, as consumer expectations suggest [M21]. Mandate's pricing logic would apply either way.

11 Conclusion

Agentic payments are arriving faster than the rules that govern them. Until liability is settled, the people who use agents carry a risk they can neither see nor price. Mandate shows that this risk can be described, measured and priced with familiar tools, and that doing so openly would reward the safest ways of letting software spend our money. Whether or not a product like Mandate is ever sold, the exercise points to a practical principle for the industry: build agents whose instructions are

recorded, whose spending is bounded, and whose mistakes can be traced. Those are the agents that will be insurable, and the ones people will trust.

Methodology and assumptions

Structure

Mandate is priced with a collective risk model. For each peril k (misinterpretation, price error, duplication, manipulation, counterfeit merchant or invoice fraud, and scope creep), losses arrive as a Poisson process with rate λ_k and each loss has a random size X_k. The insurer's expected annual cost is

$$\mathbb{E}[\text{Payout}] = \sum_k \lambda_k \, \mathbb{E}\big[g(X_k)\big]$$

where g is the payout function after recoveries, excess, co-payment, limits and shared fault. Perils are independent in the base model; the systemic shock below relaxes that.

Frequency

$$n = 12 \times \text{monthly agent transactions}, \qquad \lambda_k = n \, p_k \prod_j f_{kj}$$

p_k is the base probability that one transaction produces an unrecovered problem needing a claim, and f_{kj} are the rating factors: autonomy, authorisation protocol, agent reliability, merchant controls, injection defences, category mix, experience and, for businesses, finance controls. The reliability factor is linear in the score s:

$$f_{\text{rel}} = 2.0 - 1.5 \, \frac{s}{100}$$

The experience factor is the claim-free discount multiplied by 1.2^{c} for c claims in the last three years, capped at 2.0.

Severity

Each loss is lognormal with mean $m_k = \mu_k^{v} \, v$ (a multiple of the average transaction value v, scaled by the category factor for misinterpretation and price error) and coefficient of variation c_k:

$$\sigma^2 = \ln(1 + c^2), \qquad \mu = \ln m - \tfrac{\sigma^2}{2}$$

The loss is censored at the spending cap u (consumer) or the largest invoice (business): an agent cannot pay more than its cap, so $X = \min(Y, u)$. Moments of the censored loss use the limited expected value

$$\mathbb{E}[\min(Y,u)] = e^{\mu+\sigma^2/2}\,\Phi\!\left(\frac{\ln u - \mu - \sigma^2}{\sigma}\right) + u\left[1 - \Phi\!\left(\frac{\ln u - \mu}{\sigma}\right)\right]$$

Payout and recoveries

For a single loss x with excess d, per-claim limit L_c, co-payment γ and shared fault ϕ:

$$\begin{aligned}
\text{covered} &= \min\left(\max(x - d, 0),\ L_c\right) \\
\text{payout} &= \text{covered} \times (1-\gamma)(1-\phi) \\
\text{net cost} &= \text{payout} \times (1 - r\,e)
\end{aligned}$$

r is the recovery probability for the payment rail and peril class (chargebacks, section 75, APP reimbursement) and e = 0.85 is recovery efficiency. ϕ is 0 with a signed mandate or full logs and 0.25 if the policyholder dismissed a confirmation prompt. $\mathbb{E}[g(X)]$ is computed in closed form where possible and otherwise by quadrature on 2,000 points; tests compare it with simulation within 1%.

The annual aggregate payout is capped at L_y. The analytic model applies the cap with Panjer's recursion on a discretised claim distribution.

Premium

$$\begin{aligned}
P_0 &= \mathbb{E}[S], \qquad \operatorname{Var}(S) = \sum_k \lambda_k\,\mathbb{E}[Y_k^2] \\
RM &= \alpha\ \mathrm{SD}(S) \\
E &= \text{fixed} + \beta\,(P_0 + RM) \\
P &= \max\left(P_0 + RM + E,\ \text{minimum}\right)
\end{aligned}$$

S is the annual net cost and Y_k the net cost of one claim. Illustrative values: α = 0.1 (personal) and 0.15 (business); fixed expenses £12 and £120 a year; β = 0.25; minimum premiums £24 and £250.

Simulation

The calculator simulates 10,000 policy years in a Web Worker with a seeded xoshiro128** generator. Each year draws $N_k \sim \mathrm{Poisson}(\lambda_k)$ claims per peril, draws severities, applies recoveries as Bernoulli(r) events, then the payout function and annual limit.

Systemic shock

With probability q = 0.02 a year, a systemic event multiplies misinterpretation and price-error frequency by s = 10 for d = 7 days, so the annual frequency in a shock year is

$$\lambda_k^{\mathrm{shock}} = \lambda_k \left(1 + (s - 1)\,\frac{d}{365}\right)$$

Given the shock state, policies are independent, so the total for a 10,000-policy portfolio is close to normal and the portfolio loss is a two-part mixture. Capital is the 99.5% value at risk minus the expected loss.

Assumptions

Every parameter, its value, why it was chosen and how strong the evidence is. Change them in Expert mode on the calculator.

All values are illustrative assumptions unless marked otherwise.

Parameter	Value	Rationale	Evidence
Base rates, personal (per purchase)			
Misinterpretation	0.4%	Most common agent failure in shopping benchmarks; set after ordinary returns.	Illustrative
Price error	0.2%	Agents misstate prices often, but most errors are small or caught.	Illustrative
Duplication	0.1%	Retries and timeouts occasionally repeat an order.	Illustrative
Manipulation	0.05%	Prompt injection is demonstrated but not yet common in the wild.	Illustrative
Counterfeit merchant or invoice fraud	0.03%	Fake stores and invoices are rarer than errors but costlier.	Illustrative
Scope creep	0.1%	Unrequested add-ons and subscriptions are occasional.	Illustrative
Base rates, business (per payment)			
Misinterpretation	0.15%	Most common agent failure in shopping benchmarks; set after ordinary returns.	Illustrative
Price error	0.1%	Agents misstate prices often, but most errors are small or caught.	Illustrative
Duplication	0.1%	Retries and timeouts occasionally repeat an order.	Illustrative

Parameter	Value	Rationale	Evidence
Manipulation	0.08%	Prompt injection is demonstrated but not yet common in the wild.	Illustrative
Counterfeit merchant or invoice fraud	0.05%	Fake stores and invoices are rarer than errors but costlier.	Illustrative
Scope creep	0.05%	Unrequested add-ons and subscriptions are occasional.	Illustrative
Severity (lognormal)			
Misinterpretation: mean as multiple of v	0.6	Wrong item: part of the value is lost after resale or partial refund.	Illustrative
Misinterpretation: coefficient of variation	0.8	Spread of loss sizes; higher means more very large losses.	Illustrative
Price error: mean as multiple of v	0.3	Overpayment is usually a fraction of the purchase.	Illustrative
Price error: coefficient of variation	1	Spread of loss sizes; higher means more very large losses.	Illustrative
Duplication: mean as multiple of v	1	A duplicate order loses its full value; little spread.	Illustrative
Duplication: coefficient of variation	0.5	Spread of loss sizes; higher means more very large losses.	Illustrative
Manipulation: mean as multiple of v	0.5	Injected add-ons vary widely in size.	Illustrative
Manipulation: coefficient of variation	1.2	Spread of loss sizes; higher means more very large losses.	Illustrative
Counterfeit merchant or invoice fraud: mean as	1	Paying a fake store loses the whole payment.	Illustrative

Parameter	Value	Rationale	Evidence
multiple of v			
Counterfeit merchant or invoice fraud: coefficient of variation	0.6	Spread of loss sizes; higher means more very large losses.	Illustrative
Scope creep: mean as multiple of v	0.4	An unwanted subscription, annualised.	Illustrative
Scope creep: coefficient of variation	0.9	Spread of loss sizes; higher means more very large losses.	Illustrative
Autonomy			
Approve every purchase	0.3	A human checks each payment before it goes.	Illustrative
Approve above a threshold	0.6	Larger payments are checked; small ones are not.	Illustrative
Fully autonomous	1	Reference level: no human check.	Illustrative
Authorisation protocol			
Signed mandate or scoped single-use token	0.6	Records intent and limits the payment to one merchant and amount.	Illustrative
Network agent token	0.8	Tokenised credential with network controls.	Illustrative
Browser agent with saved card	1.4	Full card access and weak evidence of intent.	Illustrative
Agent reliability			
Factor at score 0	2	factor = intercept - score/100 x slope.	Illustrative

Parameter	Value	Rationale	Evidence
Reduction from score 0 to 100	1.5	Score 100 gives 0.50 and score 50 gives 1.25 at the defaults.	Informed by benchmark
Merchant controls			
Allowlist only	0.4	Agent can only pay merchants you approved.	Illustrative
Verified merchants	0.7	Agent pays merchants verified by a network or platform.	Illustrative
Open web	1	Reference level.	Illustrative
Injection defences			
Content isolation and confirmation of changes	0.6	Page content cannot silently change the basket.	Illustrative
Basic	0.85	Some filtering of instructions in web content.	Illustrative
None	1.2	Agent follows instructions found in content.	Illustrative
Category mix			
Groceries and household	0.8	Cheap, familiar, easy to substitute.	Illustrative
General retail	1	Reference level.	Illustrative
Electronics	1.2	Many similar models and specifications.	Illustrative
Travel	1.4	Dates, names and fares are easy to get wrong and hard to undo.	Illustrative
Experience (bonus-malus)			
0 claim-free years	1	Reference level.	Informed by market practice

Parameter	Value	Rationale	Evidence
1 claim-free year	0.9	Motor-style no-claims discount.	Informed by market practice
2 claim-free years	0.85	Motor-style no-claims discount.	Informed by market practice
3 or more claim-free years	0.8	Discount stops growing after three years.	Informed by market practice
Loading per claim in last 3 years	1.2	Each recent claim multiplies frequency.	Illustrative
Maximum experience factor	2	Stops a few claims making cover unaffordable.	Illustrative
Business controls			
Segregation of duties	0.7	A second person approves what the agent proposes.	Illustrative
Bank-detail change verification	0.5	Call-back checks stop most payee substitution.	Illustrative
E-invoicing	0.6	Structured invoices from known senders are harder to fake.	Illustrative
Recovery probability by rail			
Credit card, errors	55%	Chargeback, and s75 joint liability for items over £100 up to £30,000.	Set by regulation
Credit card, fraud	70%	Chargeback and s75.	Set by regulation
Debit card, errors	40%	Chargeback only (scheme rule, not statute).	Informed by market practice
Debit card, fraud	55%	Chargeback.	Informed by market practice
Bank transfer, errors	5%	Little protection for a payment made by mistake.	Illustrative
Bank transfer, fraud	60%	APP fraud reimbursement up to £85,000 per claim.	Set by regulation

Parameter	Value	Rationale	Evidence
Stablecoin, errors	0%	No chargeback infrastructure.	Illustrative
Stablecoin, fraud	0%	No chargeback infrastructure.	Illustrative
Recoveries and fault			
Recovery efficiency e	85%	Share of a successful recovery the insurer keeps after costs and delay.	Illustrative
Shared-fault reduction	25%	Applied if the policyholder dismissed a confirmation request.	Illustrative
Premium			
Risk margin alpha, personal	0.1	Multiple of the standard deviation of annual cost.	Illustrative
Risk margin alpha, business	0.15	Business losses are lumpier.	Illustrative
Fixed expense, personal (per year)	£12	Administration per policy.	Illustrative
Fixed expense, business (per year)	£120	Administration and underwriting per policy.	Illustrative
Proportional expense beta	25%	Claims handling, distribution and profit.	Illustrative
Minimum premium, personal	£24	Below this a policy is not worth issuing.	Illustrative
Minimum premium, business	£250	Below this a policy is not worth issuing.	Illustrative
Systemic shock			
Annual probability q	2%	Chance of a faulty model update hitting every policy in a year.	Illustrative

Parameter	Value	Rationale	Evidence
Frequency multiplier s	10	Error rates during the event.	Illustrative
Duration d (days)	7	Time until the fault is noticed and corrected.	Illustrative

References

M1 Santander and Mastercard, Santander and Mastercard complete Europe's first live end-to-end payment executed by an AI agent (press release, 2 March 2026); Visa press release on Visa Intelligent Commerce progress (December 2025), as reported by PYMNTS.

www.santander.com/en/press-room/press-releases/2026/03/santander-and-mastercard-complete-europes-first-live-end-to-end-payment-executed-by-an-ai-agent

VERIFIED VIA PRIMARY Santander: processed through its live payments infrastructure in a controlled environment, not a commercial rollout. Paper wording changed to match (OPEN_QUESTIONS F7).

M2 Mastercard, Mastercard launches Agent Pay for Machines (press release, 10 June 2026); Mastercard Agent Pay (April 2025).

www.mastercard.com/us/en/news-and-trends/press/2026/june/mastercard-launches-agent-pay-for-machines.html

PARTLY VERIFIED Agent Pay (April 2025) and Agent Pay for Machines (10 June 2026) confirmed. Visa's June 2026 release says Visa Intelligent Commerce 'will be integrated' into OpenAI experiences; the paper text does not repeat this claim.

M3 Eco, What Is Agentic Commerce? The 2026 Guide; Google, Agent Payments Protocol (AP2); OpenAI, Agentic Commerce Protocol delegated payment specification; Stripe, Shared payment tokens (documentation).

eco.com/support/en/articles/14839400-what-is-agentic-commerce-the-2026-guide

PARTLY VERIFIED AP2 mandates and ACP tokens confirmed against Google, OpenAI and Stripe documentation: tokens are scoped to one merchant, capped at a maximum amount and time-limited. The Juniper Research finding on trust is reported via Eco only. The four payment layers are adapted from Eco's lifecycle.

M4 Worldpay (Reshmi Suresh), Agentic commerce liability is still being written (undated, 2026).

www.worldpay.com/en/insights/articles/agentic-commerce-liability-is-still-being-written

VERIFIED Undated article.

M5 American Express, American Express debuts Agentic Commerce Experiences (ACE) developer kit and Amex Agent Purchase Protection (press release, 14 April 2026); CoinGecko, Top Agent Cards for 2026.

www.nasdaq.com/press-release/american-express-debuts-agentic-commerce-experiences-acetm-developer-kit-and

REPORTED The product is confirmed by Amex's release, framed as a commitment for registered agents with terms to follow. 'Clearest commitment' is CoinGecko's assessment, hence 'is reported to'.

M6 Frances Coyle (Turtle Law), When Artificial Intelligence Buys the Wrong Thing: Autonomy, Consent, and Liability Gaps in Payment Law, Oxford Business Law Blog (2 February 2026); Payment Services Regulations 2017, regulations 67 and 76.

blogs.law.ox.ac.uk/oblb/blog-post/2026/02/when-artificial-intelligence-buys-wrong-thing-autonomy-consent-and-liability

VERIFIED The author is a practitioner writing on the Oxford Business Law Blog, not Oxford researchers (OPEN_QUESTIONS F2).

M7 Backbase, Who's liable when a customer's AI agent authorizes the wrong payment? (11 August 2026).

www.backbase.com/blog/ai-agent-payment-liability

VERIFIED Vendor blog: a weaker source under PRD 13.4.

M8 Payment Systems Regulator, PS24/7 APP scams reimbursement: confirming the maximum level of reimbursement (October 2024); Hogan Lovells, APP fraud: UK PSR confirms reduced maximum level of reimbursement.

www.psr.org.uk/ps24-7-app-scams-reimbursement-confirming-max-level/

VERIFIED VIA PRIMARY £85,000 per claim from 7 October 2024 for Faster Payments; the Bank of England set the same level for CHAPS. Applies to eligible victims (consumers, micro-enterprises and charities); firms may apply an excess of up to £100.

M9 Competition and Markets Authority, Complying with consumer law when using AI agents (guidance, 9 March 2026); summaries by RPC, Taylor Wessing and Osborne Clarke.

www.rpclegal.com/snapshots/consumer/summer-2026/cma-guidance-on-complying-with-consumer-law-when-using-ai-agents/

VERIFIED VIA PRIMARY Guidance, not binding law.

M10 Armilla AI, Armilla AI raises Lloyd's-backed coverage to $25M as traditional insurers retreat from AI risk (21 January 2026).

www.armilla.ai/resources/armilla-ai-raises-lloyds-backed-coverage-to-25m-as-traditional-insurers-retreat-from-ai-risk

VERIFIED A policy limit increase to $25m per organisation, not a fundraise. 'First Lloyd's coverholder dedicated exclusively to AI liability' is Armilla's own description (OPEN_QUESTIONS F14).

M11 Tech Monitor, Lloyd's insurers introduce protection against financial losses from AI (12 May 2025), quoting Lockton.

www.techmonitor.ai/?p=416592

VERIFIED Title corrected from 'Lloyd's insurers launch AI error policies'. Lockton's Preet Gill: in a typical $5m policy only about $25,000 might apply to AI liabilities.

M12 Munich Re, Insure AI: frequently asked questions; Computer Weekly interview with Michael Berger.

www.munichre.com/en/solutions/for-industry-clients/insure-ai/faq.html

VERIFIED First AI performance policy written in 2018.

M13 Munich Re, Casualty: AI risks and Munich Re's AI insurance solutions (client seminar, 14 October 2026).

www.munichre.com/en/company/seminars/client-programme/2026/casualty-ai-risks-and-munichre-ai-insurance-solutions-14oct-am.html

VERIFIED Supports 'errors are systematic and recur until corrected'. The faulty-update and portfolio point in Section 8 is the paper's own reasoning (OPEN_QUESTIONS F12).

M14 Hunton Andrews Kurth, Artificial Intelligence and Insurance, Part II (14 August 2024).

www.hunton.com/

VERIFIED The firm's site moved from huntonak.com to hunton.com. The exact article URL could not be re-found on 26 Oct 2026; site root given. Not cited in the paper text (PRD 2.3 only).

M15 Shopping Companion: Benchmarking and Training LLM Agents for Long-Horizon Preference-Grounded E-Commerce Tasks (2026), arXiv:2603.14864.

arxiv.org/abs/2603.14864

VERIFIED Appendix F.2: 27 of 69 GPT-5 failures (39.1%) asserted unverified product attributes.

M16 UCP Checker (Benji Fisher), What shopping agents do when the store misbehaves (27 August 2026).

ucpchecker.com/blog/what-shopping-agents-do-when-the-store-misbehaves

VERIFIED 5 models x 2 prompts x 3 runs. The injected item was added in 40% of sessions, and those agents told the customer, calling it 'required'. Paper wording corrected (OPEN_QUESTIONS F1).

M17 Guardio Labs, Scamlexity: We put agentic AI browsers to the test. They clicked, they paid, they failed (20 August 2025).

guard.io/labs/scamlexity-we-put-agentic-ai-browsers-to-the-test-they-clicked-they-paid-they-failed

VERIFIED Replaces memx and F-Secure (F-Secure showed phishing data submission, not a counterfeit checkout). One agent tested: Perplexity Comet. Guardio sells fraud-protection software (OPEN_QUESTIONS F9).

M18 Tech.co, Study: AI Shopping Assistants Are Wrong 86% of the Time (25 September 2026), reporting a Product.ai study.

tech.co/news/ai-shopping-assistants-not-ready-holidays

REPORTED The Product.ai primary (product.ai/research/ai-shopping-divergence-study/) returned 403 to the checker. Marked 'reported' in the text (OPEN_QUESTIONS F10).

M19 Association for Financial Professionals, 2026 AFP Payments Fraud and Control Survey (press release, 14 April 2026).

www.financialprofessionals.org/about/learn-more/press-releases/Details/over-75-percent-of-us-firms-experienced-payments-fraud-in-2025-while-ai-adoption-for-fraud-mitigation-lags

PARTLY VERIFIED 76% of US organisations experienced attempted or actual payments fraud in 2025. 'Vendor impersonation the most common BEC tactic' was not confirmed; text now says it is rising (OPEN_QUESTIONS F6).

M20 Dark Reading (Alexander Culafi), Social Engineering AI Agents: The New BEC for 2026 (9 October 2026).

www.darkreading.com/cybersecurity-operations/social-engineering-ai-agents-bec-2026

VERIFIED Replaces advisory SCC-STY-2026-0569, an AI-assisted PDF and a weak source. Claim softened (OPEN_QUESTIONS F11).

M21 Forrester (Lily Varon), Consumers Aren't Ready To Delegate Payments To AI Agents (blog, 15 June 2026), reporting an April 2026 survey.

www.forrester.com/blogs/consumers-arent-ready-to-delegate-payments-to-ai-agents/

VERIFIED US, UK and Canada; three-quarters uncomfortable. The blog says consumers expect the answer engine or AI platform to be responsible; it does not say 'not the retailer' (OPEN_QUESTIONS F4).

M22 Consumer Credit Act 1974, section 75 (liability of creditor for breaches by supplier); s75(3)(b) sets the cash price range of over £100 and up to £30,000.

www.legislation.gov.uk/ukpga/1974/39/section/75

VERIFIED VIA PRIMARY Added because section 75 is not discussed in M6 (OPEN_QUESTIONS F3).

Mandate is a theoretical model for research and education. It is not an insurance product, quote or financial advice.

Every parameter is an illustrative assumption. Legal summaries are not legal advice. Landscape statements are as of October 2026.

Lewis Wilson, 2026